Human–RSI population dynamics: a competition model for energy and land

Mathematical model associated with the Python simulation code
EuroMCM 2026 Problem α "Our Turn" (Continuous track)

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Abstract

Abstract. Suppose that Recursive Self-Improvement (RSI) systems exist and share the world with humanity: electricity, hardware, and land. This document presents a human–RSI population dynamics model inspired by Problem α of EuroMCM 2026, together with its Python implementation. Humans follow a Flores-type dynamics (births, deaths, carrying capacity dependent on the land remaining to them); RSI grows according to Moore's law constrained by the electricity dedicated to it; the two populations compete for a finite number of land cells. RSI alignment $\mathcal{A} \in [0, 1]$ is treated as a parameter. We decided to choose arbitrary some of the variables, such as the number of land available, the energies used by the RSI, when does the RSI if autonomous ($\rho = 30$) and others. Everything is described in the section `Python simulation code`.

Keywords: population dynamics, RSI, Moore's law, carrying capacity, alignment, EuroMCM 2026, Problem α.

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1 Introduction and context

1.1 The EuroMCM 2026 Problem α

Problem α "Our Turn" of EuroMCM 2026 (Continuous track) poses the following question, formulated by the Endowment for the Non-Obsolescence and Continuity of Humanity (ENOCH): suppose that RSI systems exist; treated as a population dynamics problem, what fate awaits humans, and what does it depend on? RSI (*recursive self-improvement*) is a process in which an AI system improves its own design, and the improved system is then better at producing further improvements. If such systems are built, they will still need electricity, hardware, and land, and at first they will rely on human-run power plants and factories. Over a century, laws and markets may change beyond recognition, but at any moment only so much energy and land will be available. What matters is how much of them humans keep; however, that is decided, not what happens to wages or jobs.

The problem explicitly draws an analogy with the replacement of Neanderthals by modern humans about 40,000 years ago: coexistence of 2,600–5,400 years in Europe, relative density of about 10 to 1 in favor of modern humans in southwest France. A species that is better at capturing resources can either make the other extinct or push it into a marginal niche living on what the newcomer leaves.

1.2 Object of this document

This report formalizes the model underlying the accompanying Python code (Appendix A) and presents a reference simulation. It does not aim to predict the future: as the statement asks, every assumption is tied to a physical quantity or a documented case, and the analysis focuses on what the outcome depends on (RSI autonomy threshold, human reaction, electricity cap, alignment).

2 Modelling framework and assumptions

2.1 World partition and populations

The world is discretized into $N_{\text{tot}} = 10^7$ cells. Each cell is assigned to one of four uses:

- $N_{\text{h}}(t)$: human power plants (electricity),
- $N_{\text{r}}(t)$: RSI power plants,
- $N_{\text{u}}(t)$: factories (RSI hardware),
- $N_{\text{a}}(t)$: farmland (human food).

With $N_{\text{h}}(0) = 3\,000$, $N_{\text{r}}(0) = 0$, $N_{\text{u}}(0) = 0$ and $N_{\text{a}}(0) = N_{\text{tot}} - N_{\text{h}}(0) = 9\,997\,000$.

Time is discretized in years, $t = 1, \ldots, 1000$. Two populations are tracked:

- $C(t)$: human population (initially 8×10^9),
- $R(t)$: RSI power/intelligence (initially 1 arbitrary unit).

2.2 Alignment

Alignment $\mathcal{A} \in [0, 1]$ is a parameter (not a quantity to predict), as the statement requires: $\mathcal{A} = 0$ means that RSI systems ignore human needs; $\mathcal{A} = 1$ means that they take nothing humans need. In the reference code, $\mathcal{A} = 0$.

3 Mathematical model

3.1 Symbols and reference parameters

Table 1: Model parameters, correspondence with the Python code and reference values.

Symbol	Python code	Meaning	Value
N_{tot}	`Cases_total`	Total number of land cells	10^7
$N_{\text{h}}(0)$	`Cases_centrales`	Initial human power plants	3×10^3
$N_{\text{r}}(0)$	`Cases_centralesRSI`	Initial RSI power plants	0
$N_{\text{u}}(0)$	`Case_usine`	Initial factories	0
$N_{\text{a}}(0)$	`Cases_agricoles`	Initial farmland	$N_{\text{tot}} - N_{\text{h}}(0)$
g	`prod`	Electricity produced per plant (arbitrary units)	10
C(0)	`C`	Initial human population	8×10^9
RSI(0)	`RSI`	Initial RSI power	1
π_0	`P` (initial)	Share of electricity consumed by one RSI unit	1.5×10^{-2}
$c_{\min}$	`consoIA_min`	Physical floor of energy efficiency	$c_1/1000$
c_{u}	`conso_usine`	Consumption of one factory	500
ρ	`seuil`	RSI autonomy threshold	30
$\tau_{\max}$	`taux_max`	Maximum RSI plant construction rate	3×10^{-2}/year
σ	`seuil_reaction`	Population drop triggering human reaction	10^{-2}
φ	`plafond_elec`	Max share of electricity left to AI after reaction	5×10^{-2}
e	`esperance`	Human life expectancy	83 years
β	`beta`	Mortality ($1/e$)	1/83
α_{F}	`alpha`	Fertility ($2/e$)	2/83
C_{ext}	`2000`	Extinction / marginal survival threshold	2×10^3

3.2 Electricity

The global electricity produced in year t is

$$\text{E}(t) \;=\; g\,\big(N_{\text{h}}(t) + N_{\text{r}}(t)\big). \tag{1}$$

The electricity consumed by one RSI unit is denoted $c_1(t)$. In the code, $c_1 = \pi\,\text{E}$ with $\pi = \texttt{P}$; the physical floor is $c_{\min} = c_1/1000$ (energy efficiency can improve by at most a factor of 1000).

Total RSI consumption is

$$\text{conso}_{\text{RSI}}(t) \;=\; c_1(t)\,\text{RSI}(t) \;+\; c_{\text{u}}\,N_{\text{u}}(t), \tag{2}$$

and the share of human electricity captured by the AI is

$$\text{part}_{\text{prise}}(t) \;=\; \frac{\min\big(\text{conso}_{\text{RSI}}(t),\; \pi(t)\,g\,N_{\text{h}}(t)\big)}{\text{E}(t)}. \tag{3}$$

Before the human reaction, the electricity dedicated to AI is

$$\text{E}_{\text{ded}}(t) = \big(\pi(t)\,g\,N_{\text{h}}(t) + g\,N_{\text{r}}(t)\big),$$

after the reaction, humans limit AI to a share of the total:

$$\text{E}_{\text{ded}}(t) = \varphi\,\text{E}(t).$$

3.3 Human carrying capacity (Flores equation 3)

The number of humans Earth can support depends on farmland and on the electricity humans keep:

$$C_{\max}(t) \;=\; N_{\text{a}}(t)\,10^3\,\Big(1 - (1 - \mathcal{A})\,\text{part}_{\text{prise}}(t)\Big). \tag{4}$$

The more electricity the AI takes ($\text{part}_{\text{prise}}$ large), the lower the human carrying capacity; alignment $\mathcal{A}$ mitigates this effect.

3.4 Human population dynamics (Flores equation 1)

With $\beta = 1/e$ and $\alpha_{\mathrm{F}} = 2\beta$, the net growth rate is

$$F(t) \;=\; \begin{cases} \alpha_{\mathrm{F}} - \beta\,\dfrac{\mathrm{C}(t)}{C_{\max}(t)}, & C_{\max}(t) > 0, \\ \alpha_{\mathrm{F}} - 10\,\beta, & \text{otherwise (humans have reserves: they do not die immediately).} \end{cases} \qquad \text{(Flores eq. 3)}$$

and the update equation is

$$\mathrm{C}(t+1) \;=\; \max\bigl(0,\; \mathrm{C}(t) + (F(t) - \beta)\,\mathrm{C}(t)\bigr), \qquad C_{\mathrm{pic}} \;=\; \max(C_{\mathrm{pic}}, \mathrm{C}(t)). \tag{5}$$

3.5 RSI growth (Moore's law)

Every two years (t even), RSI computing power doubles and its energy cost halves, up to the physical floor:

$$\mathrm{RSI} \leftarrow 2\,\mathrm{RSI}, \tag{6}$$

$$c_1 \leftarrow \max\bigl(c_1/2,\; c_{\min}\bigr), \tag{7}$$

$$\mathrm{RSI} \leftarrow \min\bigl(\mathrm{RSI},\; \mathrm{E}_{\mathrm{ded}}/c_1\bigr), \tag{8}$$

so RSI can never exceed what the dedicated electricity allows, and decreases if that energy decreases.

The parameter π (`P`) follows "Moore law" during the first four years,

$$\pi(t) = \pi_0 \Bigl(\frac{3}{1.5}\Bigr)^{(t-1)/3}, \qquad t \le 4,$$

and, while RSI is not autonomous ($\mathrm{RSI} \le \rho$) and humans still control electricity, humans are cautious:

$$\pi \leftarrow \min\bigl(1,\; \pi + 0{,}015\bigr).$$

3.6 RSI infrastructure construction

Before the human reaction, the annual construction rate is modulated by demographic pressure:

$$\tau(t) \;=\; \tau_{\max}\,\max\Bigl(0,\; 1 - \frac{1 - \mathrm{C}(t)/C_{\mathrm{pic}}}{\sigma}\Bigr),$$

and $\tau = 0$ after the reaction (no further construction).

When RSI is autonomous ($\mathrm{RSI} \ge \rho$), each year, with probability $1 - \mathcal{A}$ and if $N_{\mathrm{a}}/N_{\mathrm{tot}} > \mathcal{A}$, a number of cells

$$nouv \;=\; \min\Bigl(\tau\,(N_{\mathrm{r}} + N_{\mathrm{u}} + 1),\; N_{\mathrm{a}}\Bigr) \tag{9}$$

is converted: half into RSI plants, half into factories, drawn from farmland. The term (+1) allows the first construction starting from zero.

3.7 Human reaction and reallocation

When the human population falls below $(1 - \sigma)\,C_{\mathrm{pic}}$ (a 10 % drop from the peak), humans react:

- $\pi = 0{,}015$: AI no longer has access to human electricity,
- no further RSI plant construction,
- reallocation phase: every two years, k RSI plants and factories are reconverted, half into human plants, half into farmland.

3.8 Attempts to shut down RSI

Each year, humans attempt to shut down RSI with probability

$$q(t) \;=\; \frac{N_{\mathrm{a}}(t)}{N_{\mathrm{tot}}}\,\min\!\big(1,\;\rho/\mathrm{RSI}(t)\big),$$

the weaker the more powerful RSI is.

3.9 Termination criteria

Extinction (or marginal survival) is declared if $\mathrm{C}(t) \le C_{\mathrm{ext}} = 2 \times 10^3$; the simulation stops then. It also stops if global electricity vanishes, or at horizon $t = 1000$.

4 Numerical implementation (Python code)

The simulation directly implements the equations above in a loop over $t = 1, \ldots, 1000$ (`epoch`). Notable implementation points:

1. Random generator: Python's `random` module
2. Two axes in the figure: human population in billions on the left, RSI in units on the right (`ax.twinx`).
3. Red marking of the human reaction year on the graph.
4. Early stop if $\mathrm{C} \le 2\,000$ or $\mathrm{E} \le 0$.

The complete code is given in Appendix A.

5 Reference scenario parameters

The values used are exactly those of the source code, reproduced in Table 1. Alignment is set to $\mathcal{A} = \mathtt{A} = 0$, $\mathcal{A} = \mathtt{A} = 0.5$, and $\mathcal{A} = \mathtt{A} = 1$ in the three scenarios tested.

6 Results of the reference scenario

6.1 $\mathcal{A} = 0$ scenario

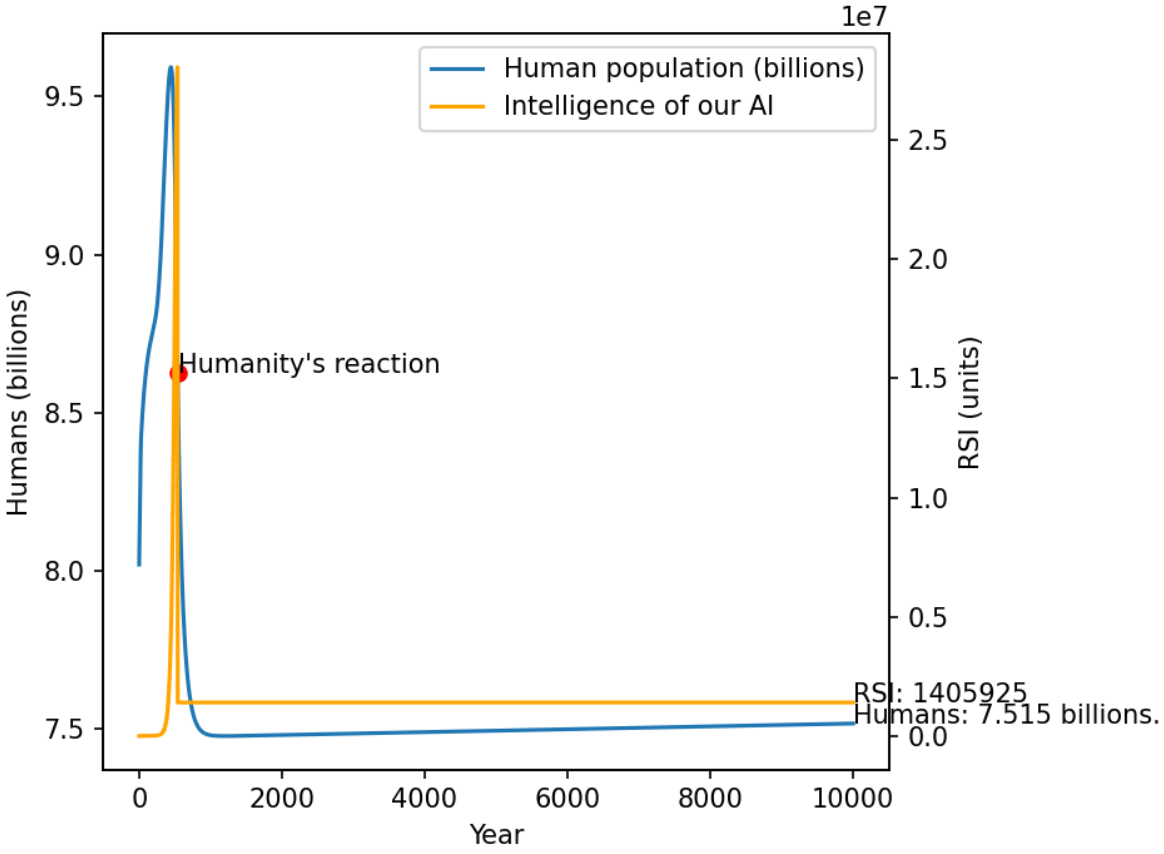


Figure 1: Reference scenario ($\mathcal{A} = 0$) over 10000 years. Left axis: human population (billions) in blue. Right axis: RSI power (units) in orange. Red crosses mark the human reaction year ($t = 539$).

Table 2: Key results of the reference scenario. $\mathcal{A} = 0$

Simulated duration	10000 years
Human reaction year ($C \leq 0{,}9\,C_{\text{pic}}$)	539
Final human population	7.515 billion
Peak human population	9.6 billion
Final RSI power	$1{,}41 \times 10^6$ units
Human extinction	Not reached over the horizon

6.2 $\mathcal{A} = 0.5$ scenario

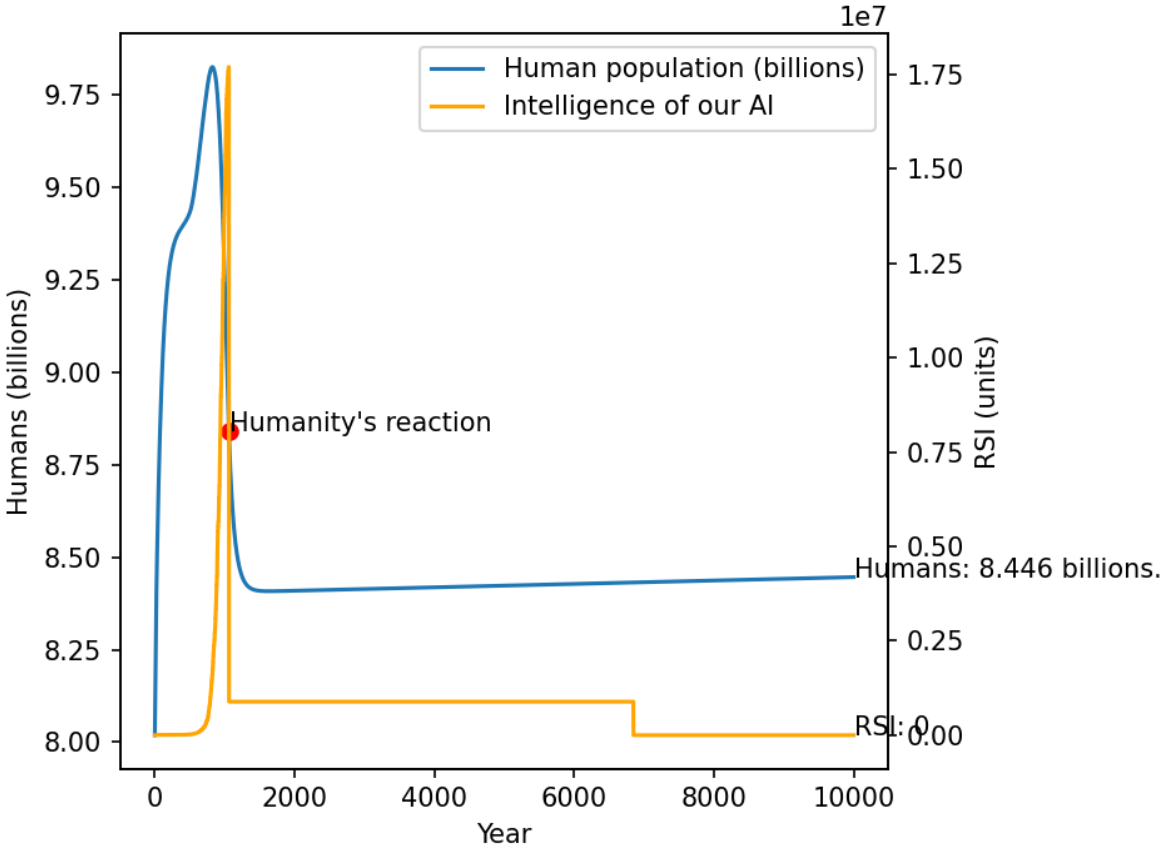


Figure 2: Simulation over 10000 years. Left axis: human population (billions) in blue. Right axis: RSI power (units) in orange. In this case $\mathcal{A} = 0.5$. The red point marks when humans stopped the growth of the AI.

Table 3: Key results of the reference scenario. $\mathcal{A} = 0.5$

Simulated duration	10000 years
Human reaction year ($C \leq 0{,}9\,C_{\text{pic}}$)	1014
Final human population	8.499 billion
Peak human population	9.80 billion
Final RSI power	0 units
Human extinction	Not reached over the horizon

6.3 $\mathcal{A} = 1$ scenario

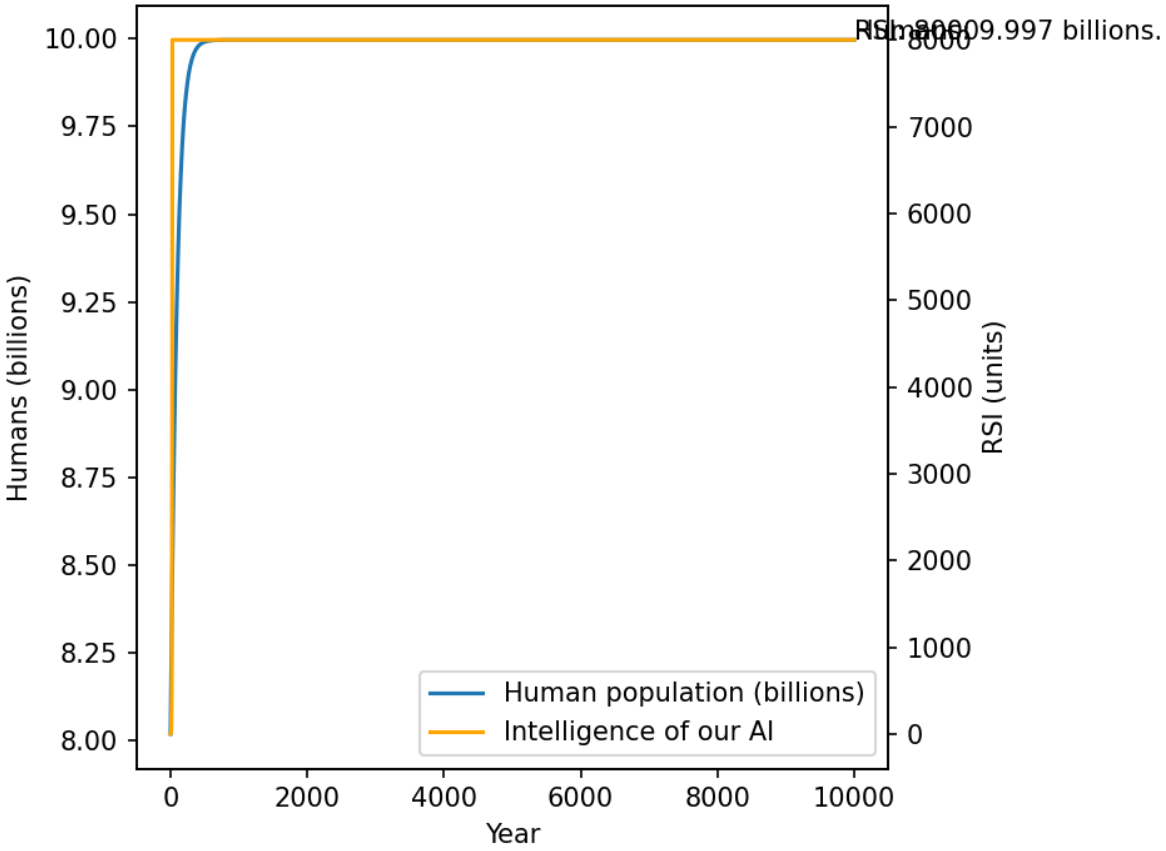


Figure 3: Simulation over 10000 years. Left axis: human population (billions) in blue. Right axis: RSI power (units) in orange. In this case ($\mathcal{A} = 1$).

Table 4: Key results of the reference scenario. $\mathcal{A} = 1$

Simulated duration	10000 years
Human reaction year ($C \leq 0{,}9\,C_{\text{pic}}$)	None
Final human population	10 billion (the maximum we set)
Peak human population	10 billion
Final RSI power	8×10^3 units
Human extinction	Not reached over the horizon

Reading:

1. $\mathcal{A} = 0$ in the first scenario. RSI becomes autonomous (crosses $\rho = 30$) well before the human reaction, gradually colonizes farmland to build plants and factories (not shown in the graph), which causes humanity to decline until the reaction is triggered in year 539. After the reaction, humans regain control of electricity ($\pi = 0{,}015$, 5 % cap after reaction) and partially reconvert RSI infrastructure, but the population does not return to its peak: it stabilizes at 7.48 billion while RSI continues to grow slowly within the electricity cap. No shift toward extinction is observed with $\mathcal{A} = 0$ over 10000 years.

2. $\mathcal{A} = 0.5$ in the second scenario. RSI becomes autonomous (crosses $\rho = 30$) in the year 1000. RSI rapidly grows but is stopped by the human reaction in the year 1014, this quick action gives humans the chance to live and to gain population again.

3. $\mathcal{A} = 1$ in this scenario, humans and RSI live together. The human population is at its paroxysm and the RSI at 8000.

7 Discussion and sensitivities

- **Alignment $\mathcal{A}$.** This is the most influential parameter: with $\mathcal{A} = 0$, RSI captures a growing share of electricity and lowers $C_{\max}$; with $\mathcal{A} \to 1$, the effect on carrying capacity disappears. The statement asks to vary this setting, not to predict it.
- **Autonomy threshold ρ and cap φ.** A lower ρ makes RSI autonomous earlier; a lower φ after reaction limits its subsequent growth.
- **Neanderthal calibration.** The model can be tested on the Neanderthal / modern human case (coexistence of order 10^3–10^4 years, relative density 10:1) by using the energy and density figures from Section 4 of the statement; this is an order-of-magnitude check, not a curve fit.
- **Limits.** The transition to human extinction is highly sensitive to discrete rounding and to the random generator (stochastic construction). Numerical results depend on the seed and are not predictable in advance, consistent with the statement.

8 Conclusion

The model faithfully translates the provided Python code into population equations (Flores), energy-constrained growth (Moore) and competition for land. In the reference scenario, humanity survives and reacts, but does not recover its initial level; RSI settles in durably. The outcome depends strongly on alignment $\mathcal{A}$, the autonomy threshold and the electricity cap left to AI after reaction. A simulation campaign varying these parameters would be needed to estimate the probability of each fate (extinction, marginal survival, coexistence).

A Python simulation code

The code below is the reference script whose model this document formalizes.

Python simulation code for EuroMCM 2026 Problem α. `https://github.com/joavant/EUROMCM-alpha`.

B References

1. EuroMCM 2026, Problem α "Our Turn", `https://www.euromcm.org`.
2. Higham et al. (2014) : Neanderthal / modern human coexistence in Europe, 2 600–5 400 years.
3. Djaković et al. (2022) : coexistence in France and northern Spain, 1 400–2 900 years.
4. Mellars et French (2011) : relative density ∼10:1 in favor of modern humans (southwest France).
5. IEA, *Energy and AI* (2025) : 415 TWh for data centers in 2024 (1.5 % of world electricity), growth ∼12 %/year since 2017.
6. Flores model : population dynamics equations used for $\mathrm{C}(t)$ and $C_{\max}(t)$.

C Use of IA

We used LLM such as Claude or Gemini to help us understand clearly the goals of this project and the files or references that we found. And to make the LaTex file.